Study report: record-leg-4

> **Record mode: replies come from the stub; quality numbers are mechanics only.**

1. Run identity

Written 2026-10-28T16:33:21.382Z from the files the appendix lists; the generator calls no model.

Run identity (source: results.json, calls.json, report-header.public.json)

Field	Value
Study	gmi-vs-agent
Run	record-leg-4
Mode	record
Task	T1
Items	5: item 1, item 2, item 3, item 4, item 5
Arms	B, C0, A, Z
Sample	1
C mode	warm
Outcome	completed
Subject	anthropic claude-haiku-4-5-20251001
Vendors called	none: the stub answers every call (record mode)
agentos	0.14.25
Harness	agentos-bench, gmi-vs-agent family (unpublished), revision 8c0361f42073
Runner	private runner at 2fdf749f5b55, CI run 40723979292
Protocol	PROTOCOL-0a.md, SHA-256 10579b4e3eb9; addendum PROTOCOL-0a-addendum.md, SHA-256 684c21c83e02

2. The question

Does a GMI on the full AgentOS runtime improve companion-style, long-horizon conversation over the agent() loop the reference product runs, and at what cost in dollars and latency per turn?

B (the reference), C0 (the GMI) and C1 (the GMI with its sentiment presets, the candidate) decide it; every other arm is a control or a diagnostic. C minus B is what the GMI runtime adds over the reference product's agent()-based path, which builds the same memory manager; B minus A is what that memory layer and the product's assembly add to agent(), since A is agent() with the reference product's prompt and tools and no memory; C minus A is the sum of the two and is reported as a description (on T1 it is read as the memory wiring check, not as a

runtime effect). A minus Z is what agent() with those tools, some of them inert, and its chain-of-thought block adds over one plain call, and Z is the floor every arm is read against.

The outcomes are GO, GO-COST, PORT, KILL, NOT SHOWN and NO VERDICT under the decision rules of the study's protocol (the Metrics and decision rules section of the study page, docs.agentos.sh/benchmarks/gmi-vs-agent). This report states what one run produced and applies no verdict rule.

3. The arms

Z is reported in every table and enters no decision rule. A minus Z, what agent() adds (its tools, the chain-of-thought block it prepends when tools are mounted, and its turn pipeline), is read on the A-against-Z pairs and the absolute scores. C against Z is read only on the absolute scores of the turns both arms ran, each arm on those turns alone (section 6); it is not measured on the axes judged only in pairs (better_reply, better_written), and no figure chains C minus A with A minus Z.

The arms (source: results.json)

Arm	Construction	In this run	Turns	Failed turns
B	The reference: the product's companion reply path as production runs it, driven by the private runner.	yes	5	0
C0	A GMI on the full AgentOS runtime with B's model, memory configuration, system text per turn and tools.	yes	5	0
A	agent() with B's static system blocks, the turn's addon, B's tools and limits, and no memory.	yes	5	0
Z	The bare loop: one streamed chat-completions call per turn over fetch, with A's model, A's static system blocks and the turn's addon, A's window and limits, and no chain-of-thought block, tools, memory or framework.	yes	5	0
B+S	B with one frozen sentence as system context: notice frustration or confusion, acknowledge it, change approach.	no	-	-
C1	C0 with the shipped sentiment presets and two mood-keyed prompt elements: the candidate.	no	-	-
A+gold	A with the gold evidence appended to the user turn (T1).	no	-	-
R	Bench retrieval over the ingested store, then a reader on the subject model (T1 anchor set).	no	-	-

4. Task families and item counts

Task families and item counts (source: results.json)

Task	Measures	In this run	Items	Turns
T1	Recall: LongMemEval-S and LOCOMO questions asked after ingestion at the datasets' dates, graded per question.	yes	5	5
T2	Persona conversations: personality, responses and in-conversation memory, with callbacks, persona facts, false premises and in-fiction pressure.	no	-	-
T3	Sentiment adaptation: the replies after frustration and confusion cues, and false-positive scenes.	no	-	-
T4	Text quality: single-turn prompts with frozen context and memory, judged in pairs.	no	-	-

5. Metrics, judges and pins

not run: no judgment set given

Flags (persona-violation-v1, adaptation-v1, false-positive-v1): a judgment is the mean of the seats' hits, and an arm's figure is the mean over its judged replies. Grades (t1-grade, probe-grade): a judgment is 1 when the final grade is correct (J1 and J2 agree, else J3, else J1) and 0 otherwise.

Pairs (pairwise-reply-v1, pairwise-text-v1): side a scores 1 for a win, 0.5 for a tie and 0 for a loss on each vote, over both orders and both seats; a mean above 0.50 favours side a. Planted controls, calibration items and the A/A control are left out of sections 6 and 7.

Intervals: 95% percentile intervals of a cluster bootstrap with 10,000 resamples (clusterBootstrap), resampling cards (T1: the question's cluster from the T1 pack).

Judges are model seats pinned by engine, model and effort; a reply whose header names another seat, engine, model or effort is discarded, and the seat runner asks again. An engine that reports no served model writes the requested slug in its header, so the model of J1 is recorded as requested and unverified, and J1 is read through its planted controls and repeats.

6. Results per family

not run: no judgment set given (the absolute judgments)

Turns and failures per arm (record mode: mechanics only) (source: results.json)

Arm	Turns	Deadline	Error chunk	Memory missing	Not the subject model	Memory call error	Other
B	5	0	0	0	0	0	0
C0	5	0	0	0	0	0	0
A	5	0	0	0	0	0	0
Z	5	0	0	0	0	0	0

7. Pairwise verdicts

not run: no judgment set given

8. Cost per arm, tokens, cached tokens and the counterfactual

not run: record mode: the stub answers every call and prices none

9. Latency per arm

Nearest-rank quantiles of the seam's timings from each turn's start; a turn cut at the deadline ranks above every completed turn and reads "deadline" where it holds the rank. Completion is the turn's end after the runner's output guard. The post-turn wait is the time a C arm waits after its reply for the work the turn queued, read from its post-turn-wait event; an arm that records none reads "-". The study's latency ceilings read the first token's p50 and p95 and the last token's p95 against B.

Latency per arm in milliseconds (record mode: mechanics only) (source: results.json)

Arm	Turns	Deadlines	First token p50	First token p95	Last token p50	Last token p95	Completion p50	Completion p95	Post-turn wait p95
B	5	0	329	375	329	375	346	474	-
C0	5	0	292	339	292	339	299	346	0
A	5	0	287	442	287	442	288	443	-
Z	5	0	305	353	305	353	305	354	-

10. The determinism gate and the stability check

not run: no determinism record given (two record runs of one script compared by T5's gate)

not run: no judgment set given (the judging checks)

The runner's error rows: each turn's failure, a call left unanswered or answered with an error status, a call from another arm's key during a turn (checked on the run's own call log), and harness errors.

Checks on this run (source: results.json)

Check	Count
Stray connections	0

11. The spend ledger

not run: no ledger given (a record run spends nothing)

12. Limitations

Record mode answers every call from the stub, so its quality figures measure the harness's mechanics and not the arms.

B runs in a private runner on the reference product's own code: section 3 describes its construction, and its code is not published. Every other arm's code is in the agentos-bench family, which is not published with this report.

Z is a control: a turn that needs a tool or a memory records Z's miss in the task's own measure, and Z enters no decision rule.

Judges are model seats; their verdicts are read through the planted controls and the judging checks, not taken as ground truth.

Intervals assume the clusters are independent; pairwise means pool both orders and both seats.

This report decides nothing: the analysis applies the decision rules of the study's protocol.

Deviation: Materials writer: Claude Code 2.1.289 with Claude Fable 5.1 at effort high, where the design named a Gemini model, which its client no longer offered on the day.

Deviation: Routing: the runner sends every vendor call of the reference arm to the stub or the metering proxy by rewriting the vendors' hosts inside its own process, since the reference product's calls take no base address of their own.

Deviation: The reference arm drives the product's streaming reply path, and its history is the last 50 stored messages the product reloads on every turn, where the other arms keep a window of 60.

Deviation: A GMI turn stops its tool loop at 5 model steps where the reference arm allows 12; the pilot report counts the reference arm's turns past 5 steps.

Deviation: The subject model was read from the reference product's reply records over the days they cover, a window shorter than the 30 days the design planned; every production setting along the reference arm's path is mirrored, beyond the list the design named.

13. Appendix: rubric hashes, materials provenance and the price table

not run: no judgment set given (the rubric hashes)

not run: no price table given (the day's prices)

No material excluded.

Inputs (source: every file this report read)

Input	File	SHA-256
results	results.json	7fad3da1077c09349e4e9f90a90ed67858b14784c33fc35a0f1e154360cbeae0
calls	calls.json	30cdc06d79ae2775c0d661759c1897f0680c66b55f77327c2382721f81625591
header	report-header.public.json	0824c42b854b27dd5f7224a880bc97babebb159fe068ea8e2e36c98e218362ad
materials	BRIEF.md	0984553b2f01690c6966e9af448cab652790f907cc235d62339c54dec9bea522

Input	File	SHA-256
materials	m01.prompt.md	c1f0dcd334483a5b6c8fd923ef68e238137d14c5f466331ae8b8d1a357b7e830
materials	m01.reply.txt	2a29ccdc1b87e1556178bb06ad4296f900406033374a5173f38a797924c97d16
T1 pack	haystack 1	a04ca369e5cab724ab65d7f85c16371bd0b93c5a46a360451828ce9afc841a49
T1 pack	haystack 2	0bf5d2ff595e4791c0b02775c80fa7a0c2c3685c17a960627fbe21841b1e2577
T1 pack	haystack 3	c0cde690ee161dd86ec63bbcac75637ec3ce6d633c06c659534884ac80d4c0ad
T1 pack	haystack 4	ebdd85f075360012569dd75c85874c04dc1adf6dc81fc243137e3a955e94c516
T1 pack	haystack 5	673bc98414fc02481be70f4fcf6658136a1358b9f5a4945869f91722b4f24379
T1 pack	haystack 6	f2f08850af4a2b747483a5dd921af2b8a8ffc9c573cd3b74e822514b08412c99
T1 pack	haystack 7	db4b45823a155be77669fa5fbe5930a409a0bdbe068d8bf1400fd0fd1f1ad224
T1 pack	haystack 8	89c082198e64aae4a3bbfefbf769b222733564bbce87158d1e2def45d319d2ef
T1 pack	haystack 9	90b3cfe022cf49ae69d4e925d2a636d4a07be9b7609c3c551e3fcfd0b509fcbf
T1 pack	haystack 10	e647ec7ad89c528428c630dcea15c01341aca6aebc5304676c2222b71689141a
T1 pack	pack.json	1085990f9b05ebfd35cdec94404a115e39d13ddc8b118a5f6ac7153e5885fde7

Materials files (source: materials)

Files	SHA-256 of the set
3	c0eb554ce1db87d06e3f0fab50cd7e4c2154b84aab9f0a17693fadd59a9387c4

Materials batches (source: materials)

Batch	Seat	Engine	Model	Effort	Items	Reply SHA-256
m01	M	claude-code 2.1.289	claude-fable-5-1	high	40	2a29ccdc1b87e1556178bb06ad4296f900406033374a5173f38a797924c97d16

T1 pack (source: T1 pack)

File	SHA-256
longmemeval-s source: longmemeval_s.json	08d8dad4be43ee2049a22ff5674eb86725d0ce5ff434cde2627e5e8e7e117894
locomo source: locomo10.json	79fa87e90f04081343b8c8debecb80a9a6842b76a7aa537dc9fdf651ea698ff4

File	SHA-256
haystack 1	a04ca369e5cab724ab65d7f85c16371bd0b93c5a46a360451828ce9afc841a49
haystack 2	0bf5d2ff595e4791c0b02775c80fa7a0c2c3685c17a960627fbe21841b1e2577
haystack 3	c0cde690ee161dd86ec63bbcac75637ec3ce6d633c06c659534884ac80d4c0ad
haystack 4	ebdd85f075360012569dd75c85874c04dc1adf6dc81fc243137e3a955e94c516
haystack 5	673bc98414fc02481be70f4fcf6658136a1358b9f5a4945869f91722b4f24379
haystack 6	f2f08850af4a2b747483a5dd921af2b8a8ffc9c573cd3b74e822514b08412c99
haystack 7	db4b45823a155be77669fa5fbe5930a409a0bdbe068d8bf1400fd0fd1f1ad224
haystack 8	89c082198e64aae4a3bbfefbf769b222733564bbce87158d1e2def45d319d2ef
haystack 9	90b3cfe022cf49ae69d4e925d2a636d4a07be9b7609c3c551e3fcfd0b509fcbf
haystack 10	e647ec7ad89c528428c630dcea15c01341aca6aebc5304676c2222b71689141a
pack.json	1085990f9b05ebfd35cdec94404a115e39d13ddc8b118a5f6ac7153e5885fde7